

Student A: Diya Jain

Student B: Suraj Kumar

Section 1: Problems for Student A

Q1-)

A)

```
if(g<=h)
    g=(g+h)/128;
else
    h= (g-h)*12;
```

RISC-V code:

Student A's draft answer:

bge t1, t0 Then

Else:

```
sub    t2, t0, t1
slli   t3,t2,3
slli   t4,t2,2
add    t5,t3,t4
add    t1, t5, 0
j      Endif
```

Then:

```
add    t0,t0,t1
srli   t0,t0,7
```

Endif:

Student B's comments:

#if t0 is less than or equal to t1 then go to Then

```
#t2= t0-t1
#t3= t2<<3
#t4= t2<<2
#t5 = t3 +t4
#t1= t5+0
#jump to endif
```

```
#t0= t0+1
#t0= t0>>7
```

#done

Judgement: The code is correct and clear.

Discussion summary: This is an IF-ELSE assembly programming question.

Student A's final answer:

bge t1, t0 Then

Else:

```

sub    t2, t0, t1
slli   t3,t2,3
slli   t4,t2,2
add    t5,t3,t4
add    t1, t5, 0
j Endif

```

Then:

```

add    t0,t0,t1
srli   t0,t0,7

```

Endif:

B)

```

if(g>=h)
    g=g+h/2;
else
    g=h*30;

```

RISC-V code:

Student A's draft answer:

Student B's comments:

bge t0,t1 Then

#if t0 is less than or equal to t1 then goto Then

Else:

```

slli   t2,t1,5
slli   t3,t1,1
sub    t1, t2,t3

```

```

#t2= t1<<5
#t3= t1<<1
#t1=t2-t3

```

Then:

```

srli   t1,t1,2
Add    t0,t0,t1

```

```

#t1= t1>>2
#t0= t0+t1
#done

```

Judgement: The code is correct and clear.

Discussion summary: This is an IF-ELSE assembly programming question.

Student A's final answer:

bge t0,t1 Then

Else:

```

slli   t2,t1,5

```

```
slli    t3,t1,1
sub     t1, t2,t3
```

Then:

```
srli    t1,t1,2
Add     t0,t0,t1
```

Q2-)

Student A's draft answer:

Student B's draft answer:

a-) 29

```
addi S7, 0, 29
```

#S7=0+29

b-) -214

```
addi S7, 0, -214
```

#S7=0+(-214)

c-) 0xEEEEEEFAB

12th bit -> F -> 1111

Adding 1:

0xEEEEEE + 1 = 0xEEEEEF

```
lui     S7, 0xEEEEEF
addi    S7,S7, 0xFAB
```

d-) 0xEEEEEEFAB

12th bit -> 1 -> 0001

```
lui     S7, 0xEDCBA
ori     S7,S7, 0x123
```

Judgment: The code is correct and clear to understand.

Discussion Summary: one line statements to give out the desired output.

Student A's final answer

a-) 29

addi S7, 0, 29

b-) -214

addi S7, 0, -214

c-) 0xEEEEEFAB

12th bit -> F -> 1111

Adding 1:

0xEEEEEE + 1 = 0xEEEEEF

```
lui    S7, 0xEEEEF
addi   S7, S7, 0xFAB
```

d-) 0xEEEEEFAB

12th bit -> 1 -> 0001

```
lui    S7, 0xEDCBA
ori    S7, S7, 0x123
```

Q3.

A)

$A[3] = A[1] + 7 * A[2]$
RISC-V code

Student A's draft answer:

```
lw t0, 4(a0)
lw t1, 8(a0)
slli t2, t1, 3
sub t2, t2, t1
add t2, t0, t2
sw t2, 12(a0)
```

Student B's comments:

```
#t0 = A[1]
#t1 = A[2]
#t2 = (A[2] << 3) = 8 * A[2]
#t2 = 7 * A[2]
#t2 = A[1] + 7 * A[2]
#A[3] = A[1] + 7 * A[2]
#done
```

Judgement: The code is correct and clear

Discussion Summary: the code does the array arithmetic and stored the final value in A[3]

Student A's final answer:

```
lw t0, 4(a0)
```

```
lw t1, 8(a0)
slli t2, t1, 3
sub t2, t2, t1
add t2, t0, t2
sw t2, 12(a0)
```

B)

$A[i] = A[i-1] + A[i-2]$
RISC-V code

Student A's draft answer:

```
slli t1, t0, 2
add t1, t1, a0
lw t2, -4(t1)
lw t3, -8(t1)
add t2, t2, t3
sw t2, 0(t1)
```

Student B's comments:

```
#t1 = (i << 2) = 4*i
#t1 = address of A[i]
#t2 = A[i-1]
#t3 = A[i-2]
#t2 = A[i-1] + A[i-2]
#A[i] = A[i-1] + A[i-2]
#done
```

Judgment: The code is correct and clear to understand.

Discussion Summary: the code does the array arithmetic and stored the final value in A[i]

Student A's final answer:

```
slli t1, t0, 2
add t1, t1, a0
lw t2, -4(t1)
lw t3, -8(t1)
add t2, t2, t3
sw t2, 0(t1)
```

Q4.)

a-) if (str[i] < '0' || str[i] > '9')
count+=1;

Student A's draft answer:

```
lb t2, 1(t0)
addi t3, zero, '0'
addi t4, zero, '9'
blt t2, t3, Do
bge t2, t4, Do
j exit
```

Student B's comments:

```
# loading t2 to [t0+1]
#t3 = '0'
#t4 = '9'
#if t2 is less than t3 then goto Do
#if t2 is greater than or equal to t4 then Do
#jump to exit
```

Do: addi t1,t1,1 #Perform t1=t1+1

Exit: #Exit
 #Done

Judgement: The code is correct and clear to understand.

Discussion Summary: using the if statement the code increases the value of count by 1.

Student A's final answer:

```
lb      t2,1(t0)
addi    t3, zero, '0'
addi    t4, zero, '9'
blt     t2,t3, Do
bge     t2,t4, Do
j       exit
```

Do: addi t1,t1,1

Exit:

```
b-) if (str [i] >='a' && str[i] <='z')
      str [i]= str[i]-32;
```

Student A's draft answer:

Student B's comments:

```
lb      t2, 1(t0)
bge     t2, t3, L0
j       Exit
```

```
# loading t2 to [t0+1]
#if t2 is greater than or equal to t3 then goto L0
# jump to exit
```

```
L0 :    ble t4, t2, Do
j       Exit
```

```
#if t4 is less than or equal to t2 then goto Do
#Jump to exit
```

Do: sub t2, t2, 32

#Perform t2= t2-32

Exit: #Exit
 # Done

Judgement: The code is correct and clear to understand.

Discussion summary: using the if statement the code increases the value of count by 1.

Student A's final answer:

```
lb      t2, 1(t0)                                 # loading t2 to [t0+1]
```

```
bge    t2, t3, L0  
      j Exit
```

```
L0 :   ble t4, t2, Do  
      j Exit
```

```
Do:    sub t2, t2, 32
```

```
Exit:
```

Section B: Suraj Kumar

Q1-)

a-)

```
If (g<=h)
    g = g+(g-h)/32;
else
    h= (g+h) * 18;
```

Student B's draft answer:

```
g= t0, h= t1
bge t1, t0, Then

Else:
    add t2,t0,t1
    slli t3,t2,4
    slli t4,t2,1
    add t5,t3,t4

Then:
    sub t2, t0, t1
    srli t3, t2, 5
    add t0, t0,t3

Endif:
```

Student A's comments:

if t0<=t1 then goto Then

```
t2= t0+t1
t3= t2<<4
t4= t2<<1
t5= t3+t4
```

```
t2= t0-t1
t3= t2>>5
t0=t0+t3
#done
```

Judgement: The code is correct and clear to understand.

b-)

Student B's draft answer:

```
blt    t0, t1, Else
slli    t2,t1,3
add     t2,t2,t1
add     t1,t2,zero
```

```
Else:
add     t2,t0,t1
srli    t2,t2,2
sub     t3,t0,t2
add     t0,t3,0
```

Student A's comments:

```
jump to Else if t0 <t1
t2= t1<<3
t2= t2+t1
t1= t2 + zero
```

```
t2= t0+t1
t2= t2>>2
t3= t0-t2
t0= t3+0
```

Judgement: The code is correct and clear to understand.

Q2-)

Solution-)

Student B's draft answer:

Student A's comments:

a-) 37

addi S7, 0, 37

b-) -306

addi S7, 0, -306

c-) 0xABCDFEDC

12th bit -> E -> 1110

Adding 1:

0xABCDF + 1 = ABCE0

lui S7, 0xABCE0

addi S7, S7, 0xEDC

d-) 0xBFDC9153

12th bit -> 0001

lui S7, 0xBFDC9

ori S7, S7, 0x153

Q3-)

a-)

$A[8] = A[6] + 11 * A[7]$

Student B's draft answer:

Student A's comments:

Let $A[7] = t0$

$A[6] = t1$

lw t0, 28(a0)

lw t1, 24(a0)

slli t2, t0, 3

slli t3, t0, 1

#loading a7 to t0

#loading a6 to t1

multiply a7 to 8

multiply a7 to 2

add	t4, t2, t3	#add t2 and t3 to t4
add	t4,t4,t0	#add t4 to 0
add	t5,t1,t4	#add t5 to t1 and t4
		#done

Judgement: The code is correct and clear to understand.

b-)

$A[i] = 3 * A[i-2] + 5 * A[i-3]$

Student B's draft answer:

Student A's comments:

slli t1,t0,2	#t1=t0<<2
add t1,t1,a0	#t1=t1+a0
lw t2, -8(t1)	#loading t2
lw t3,-12(t1)	#loading t3
slli t4,t2,1	#t4= t2<<1
add t4,t4,t2	#t4=t4+t2
slli t5,t3,2	#t5= t3<<2
add t5,t5,t3	#t5=t5+t3
add t6, t4, t5	#t6= t4+t5
sw t6, 0(t1)	#save t6
	#done

Judgement: The code is correct and clear to understand.

Q4-)

a-) if (str[i] < '0' || str[i] > '9')
count+=1;

Student B's draft answer:

Student A's comments:

lb	t2,1(t0)	# loading t2 to [t0+1]
addi	t3, zero, '0'	#t3 ='0'
addi	t4, zero, '9'	#t4='9'
blt	t2,t3, Do	#if t2 is less than t3 then goto Do
bge	t2,t4, Do	#if t2 is greater than or equal to t4 then Do
	j exit	#jump to exit
Do:	addi t1,t1,1	#Perform t1=t1+1
Exit:		#Exit
		#Done

Judgement: The code is correct and clear to understand.

b-) if (str [i] >='a' && str[i] <='z')
 str [i]= str[i]-32;

Student B's draft answer:

```
lb    t2, 1(t0)
bge   t2, t3, L0
      j Exit

L0 :   ble t4, t2, Do
      j Exit

Do:    addi t2, t2, 32

Exit:
```

Student A's comments:

```
# loading t2 to [t0+1]
# if t2 is greater than or equal to t3 then goto L0
# jump to exit

# if t4 is less than or equal to t2 then goto Do
# Jump to exit

# Perform t2= t2+32

# Exit
# Done
```

Judgement: The code is correct and clear to understand.